HYDROLEVEL

Vehicle Load Intelligence & Digital Twin

Engineering Analysis Report

Prototype / Academic Engineering Project — Team Volts and Bolts

Parameter	Value
Completed rows included	20
Valid rows in exported package	20
Minimum rows required	20
Measurement points	FL · FR · RL · RR
Screening threshold	±10.00 kg
Equalization blend	50% toward arithmetic reference
Mean total load	1111.28 kg
Pre-alert rows	19
Post-alert rows	14

Project Team

Executive Summary

HydroLevel processed 20 valid rows. Mean total measured load is 1111.28 kg. 19 rows were flagged before equalization and 14 remained flagged after the configured 50% equalization blend. Front/rear mean distribution is 46.9% / 53.1%, while left/right is 49.8% / 50.2%. Insurance review support status: REVIEWABLE; engineering screening: ANOMALIES REQUIRE REVIEW. These are project screening outputs, not an insurance payout decision.

Data provenance: imported/user-provided measurements are retained as RAW values; derived values are calculated by HydroLevel. Demo/simulated data must be identified as such outside this report.

Insurance Review Support

Item	HydroLevel result
Review status	REVIEWABLE
Engineering screening	ANOMALIES REQUIRE REVIEW
Post-alert rate	70.0%
Payout decision	NOT DETERMINED BY HYDROLEVEL

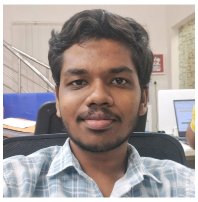
Manual review	REQUIRED
Scope	HydroLevel provides engineering evidence and screening only. Insurance coverage, claim validity and payou

Team & Responsibilities

Member	Role	Education	Responsibility	Contact
Saai Varshan S	Team Lead	B.E. Mechanical Engineering · 3rd Year	Overall project, system architecture, coordination, testing and final decisions.	saaivarshan69@gmail.com +91 93422 96487 https://www.linkedin.com/in/saai-varshan-8a62b7328
Suheerthan S	Mechanical & Vehicle Dynamics Engineer	B.E. Mechanical Engineering · 3rd Year	Vehicle model, load distribution, suspension/load concepts, mechanical design and CAD.	suheerthan2514@gmail.com +91 90433 09288 https://www.linkedin.com/in/suheerthan-s-18aa58327/
Santhosh R	Electronics & Sensor Integration Engineer	B.E. Mechanical Engineering · 3rd Year	Sensors, wiring, data acquisition, calibration and hardware testing.	santhoshravijv@gmail.com +91 97105 81763 https://www.linkedin.com/in/santhosh-ravi-95097b328

Saai Varshan S

Suheerthan S



Santhosh R

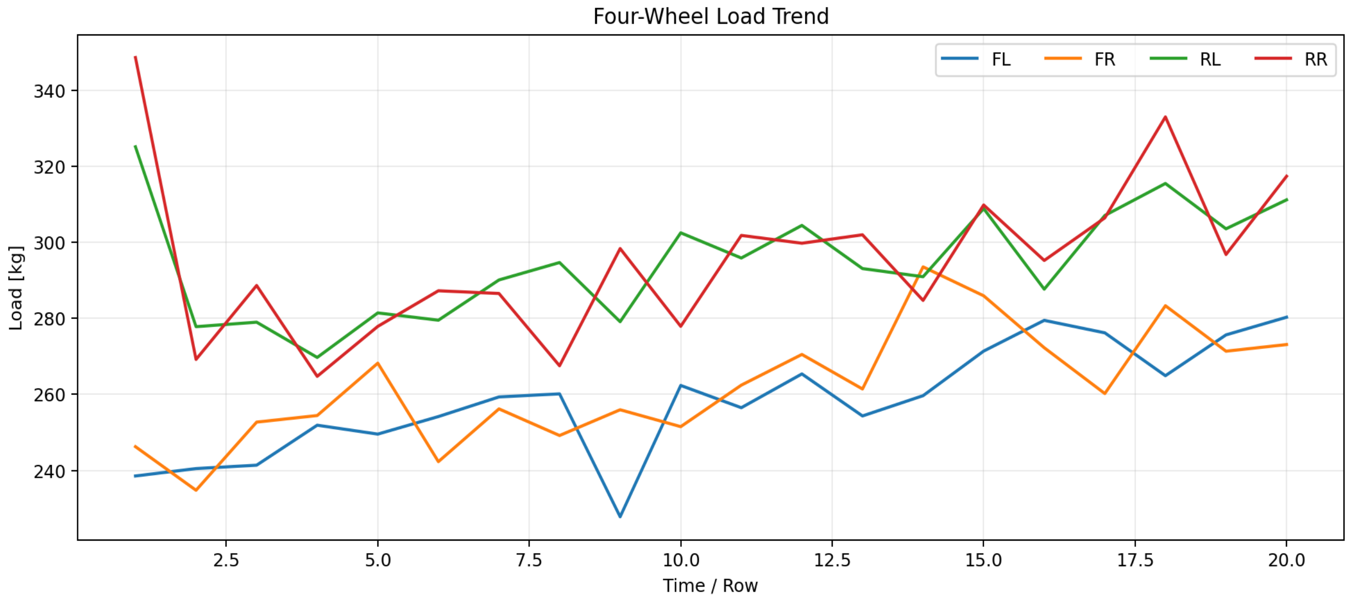
Row-by-Row Analysis

Row	FL	FR	RL	RR	Total	Avg	Pre	Post	Alerts
1	238.6	246.3	325.1	348.6	1158.5	289.6	DANGER	DANGER	FL,FR,RL,RR
2	240.5	234.8	277.8	269.2	1022.3	255.6	DANGER	WARNING	FL,FR,RL,RR
3	241.4	252.7	279.0	288.6	1061.8	265.4	DANGER	WARNING	FL,FR,RL,RR
4	251.9	254.4	269.7	264.8	1040.8	260.2	SAFE	SAFE	—
5	249.6	268.2	281.4	277.9	1077.1	269.3	WARNING	SAFE	FL,RL
6	254.2	242.3	279.5	287.2	1063.3	265.8	DANGER	WARNING	FL,FR,RL,RR
7	259.3	256.2	290.1	286.5	1092.2	273.0	WARNING	SAFE	FL,FR,RL,RR
8	260.1	249.2	294.7	267.5	1071.5	267.9	DANGER	WARNING	FR,RL
9	227.8	256.0	279.1	298.4	1061.3	265.3	DANGER	WARNING	FL,RL,RR
10	262.4	251.5	302.5	277.9	1094.3	273.6	DANGER	WARNING	FL,FR,RL
11	256.5	262.4	295.9	301.8	1116.6	279.2	DANGER	WARNING	FL,FR,RL,RR
12	265.4	270.5	304.4	299.7	1140.1	285.0	WARNING	SAFE	FL,FR,RL,RR
13	254.3	261.4	293.1	302.0	1110.8	277.7	DANGER	WARNING	FL,FR,RL,RR
14	259.7	293.5	290.9	284.7	1128.8	282.2	DANGER	WARNING	FL,FR
15	271.4	285.9	308.9	309.8	1176.0	294.0	DANGER	WARNING	FL,RL,RR
16	279.5	272.3	287.7	295.2	1134.6	283.7	WARNING	SAFE	FR,RR
17	276.2	260.2	307.0	306.4	1149.9	287.5	DANGER	WARNING	FL,FR,RL,RR
18	264.9	283.3	315.4	332.9	1196.6	299.2	DANGER	WARNING	FL,FR,RL,RR
19	275.6	271.4	303.5	296.8	1147.3	286.8	WARNING	SAFE	FL,FR,RL
20	280.3	273.1	311.1	317.4	1181.9	295.5	DANGER	WARNING	FL,FR,RL,RR

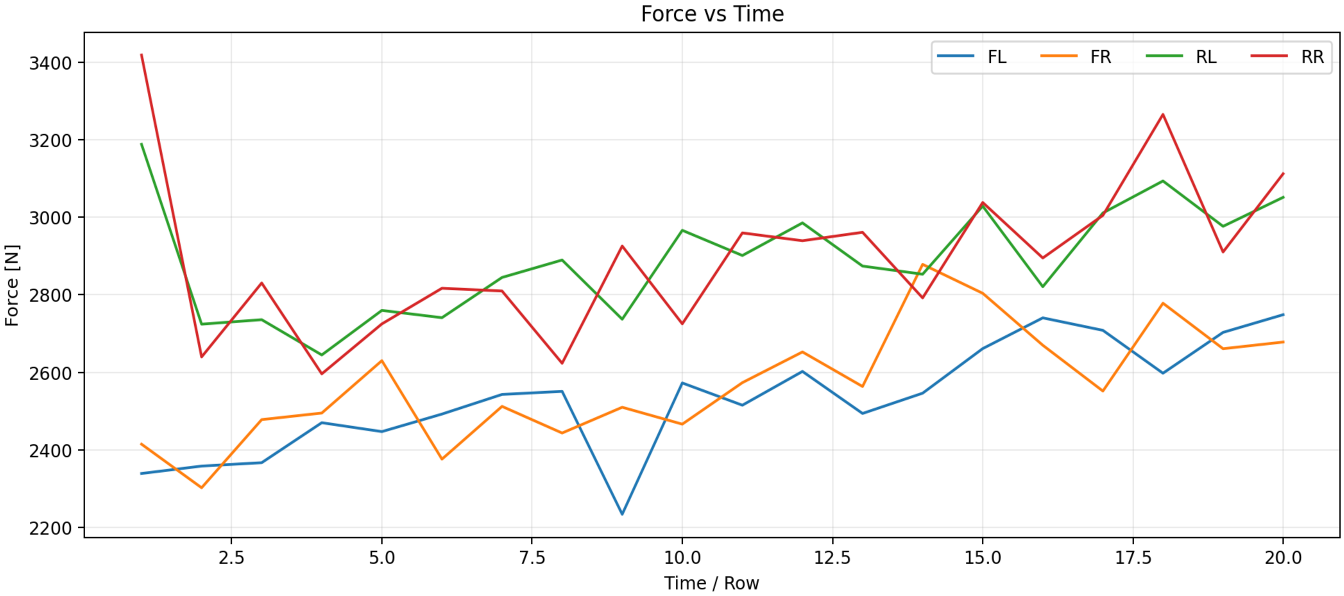
Engineering Graph Package

Graphs are generated from the same processed dataset used by the dashboard. Raw and equalized values are kept separate.

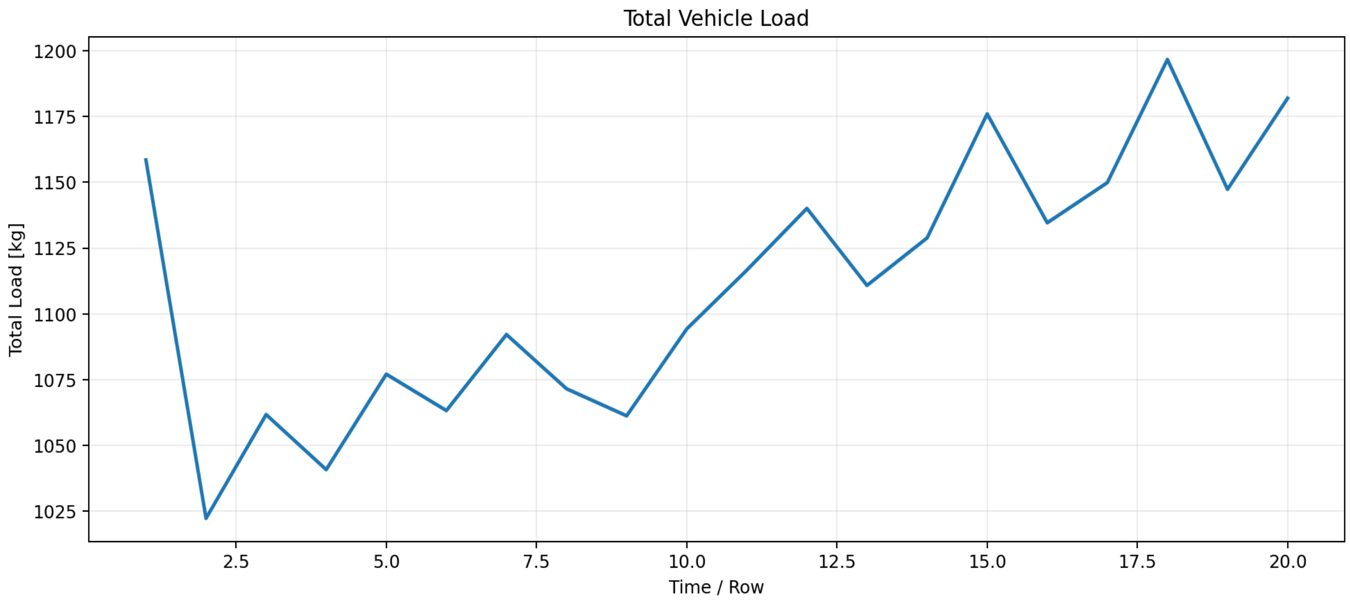
01 FOUR WHEEL LOAD TREND



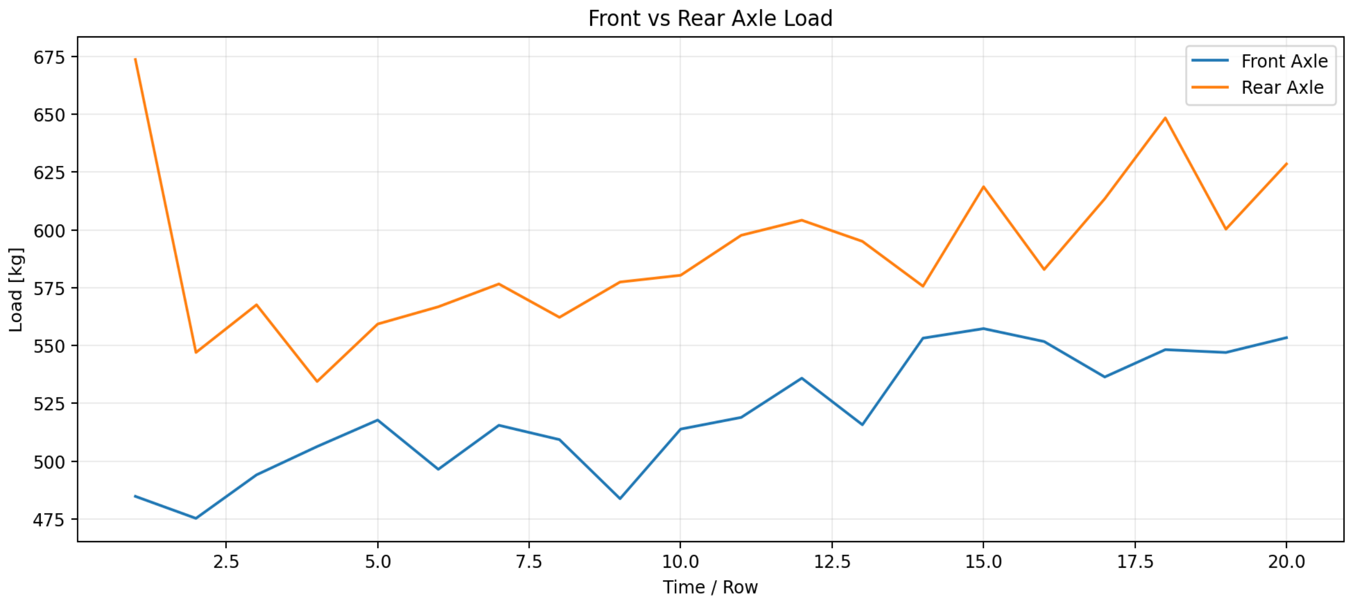
02 FORCE VS TIME N



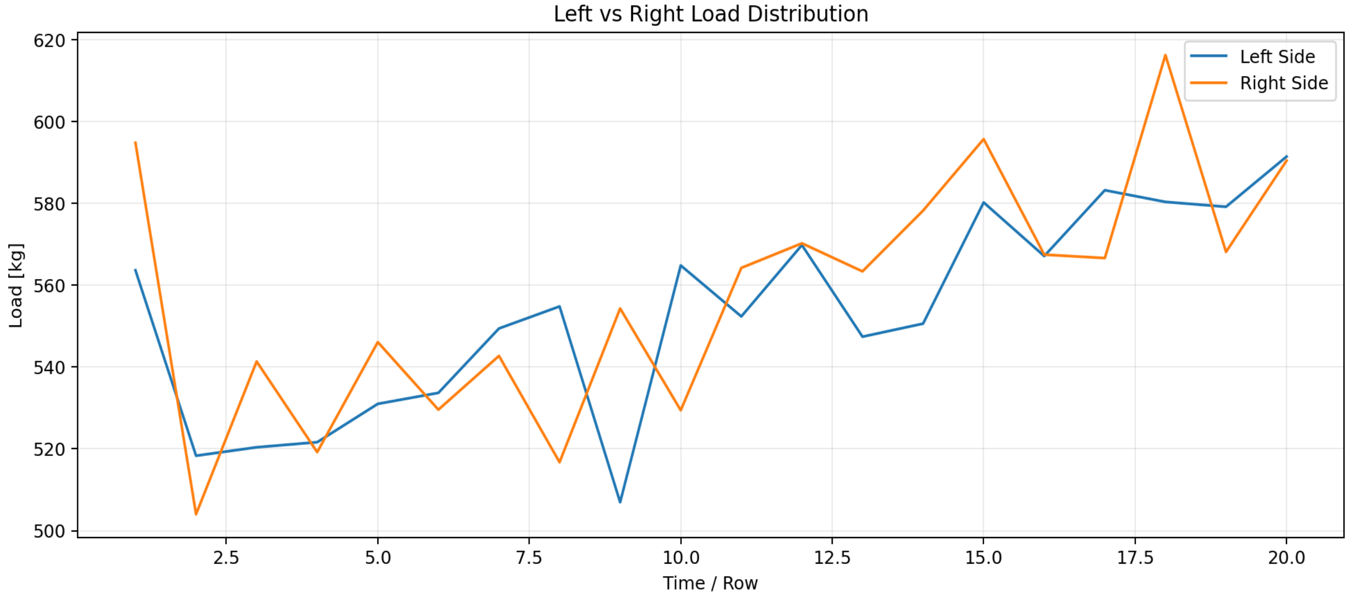
03 TOTAL VEHICLE LOAD



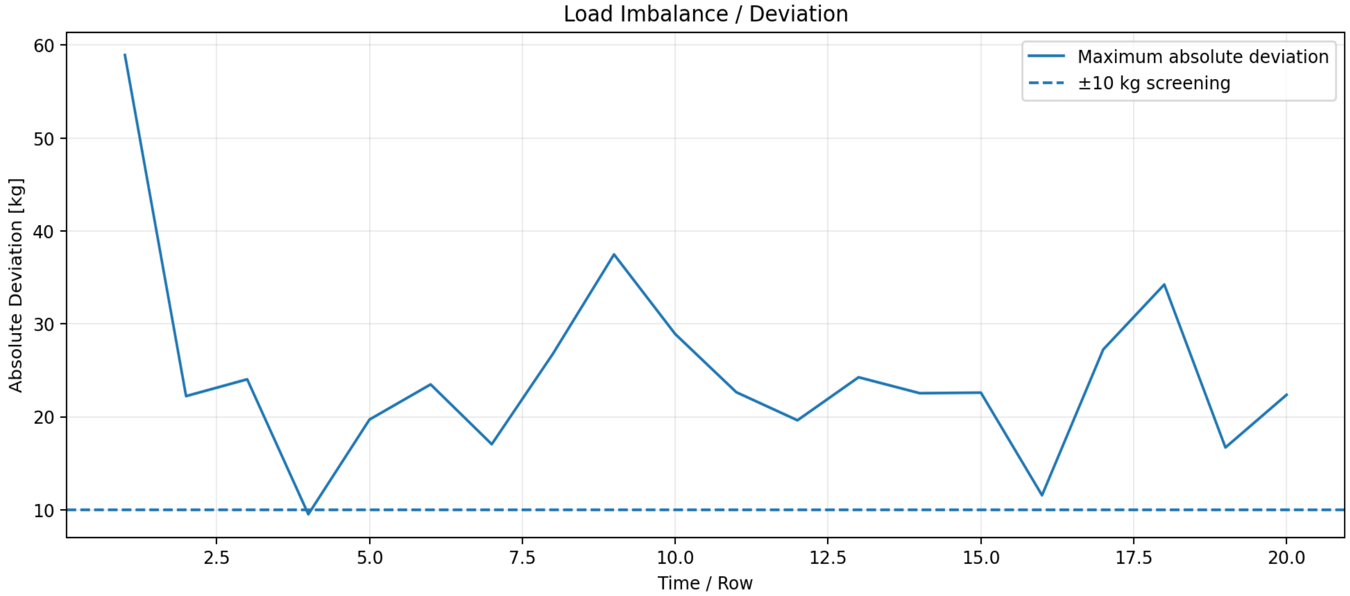
04 FRONT REAR AXLE



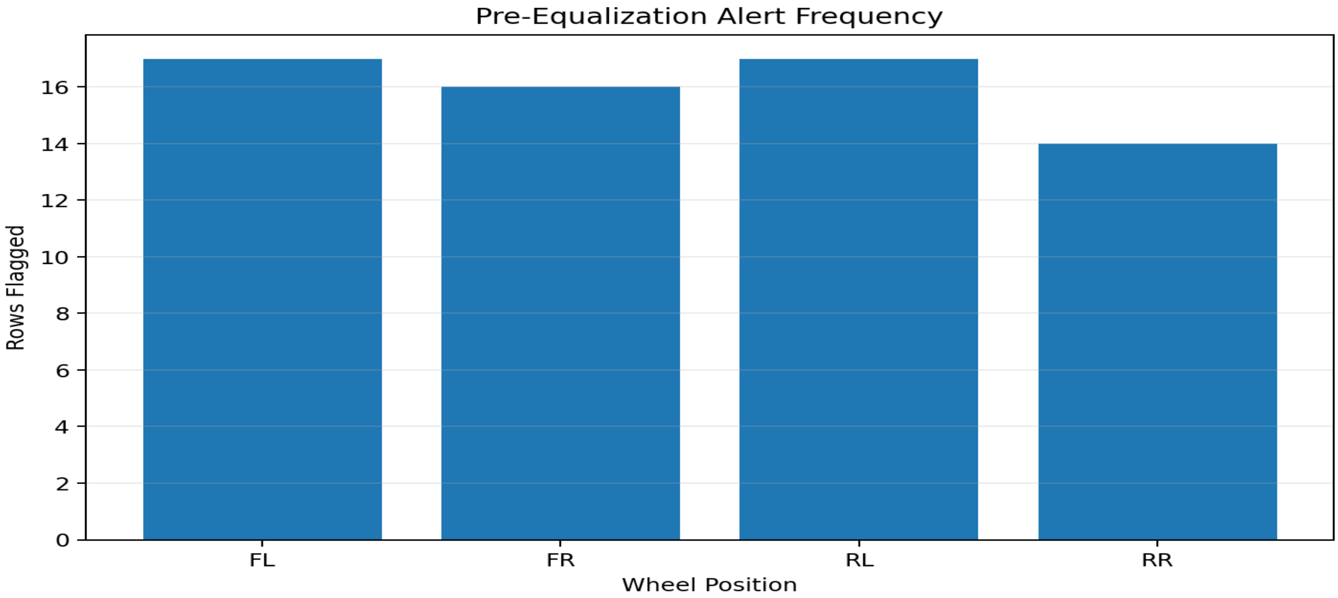
05 LEFT RIGHT DISTRIBUTION



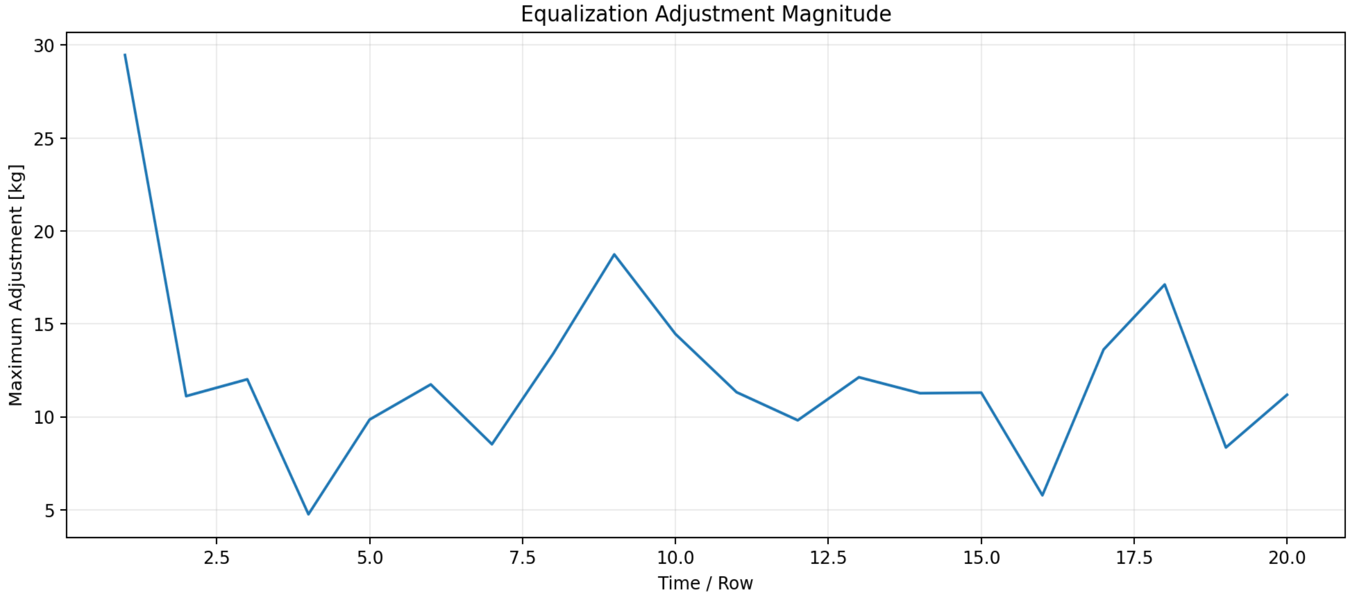
06 DEVIATION THRESHOLD



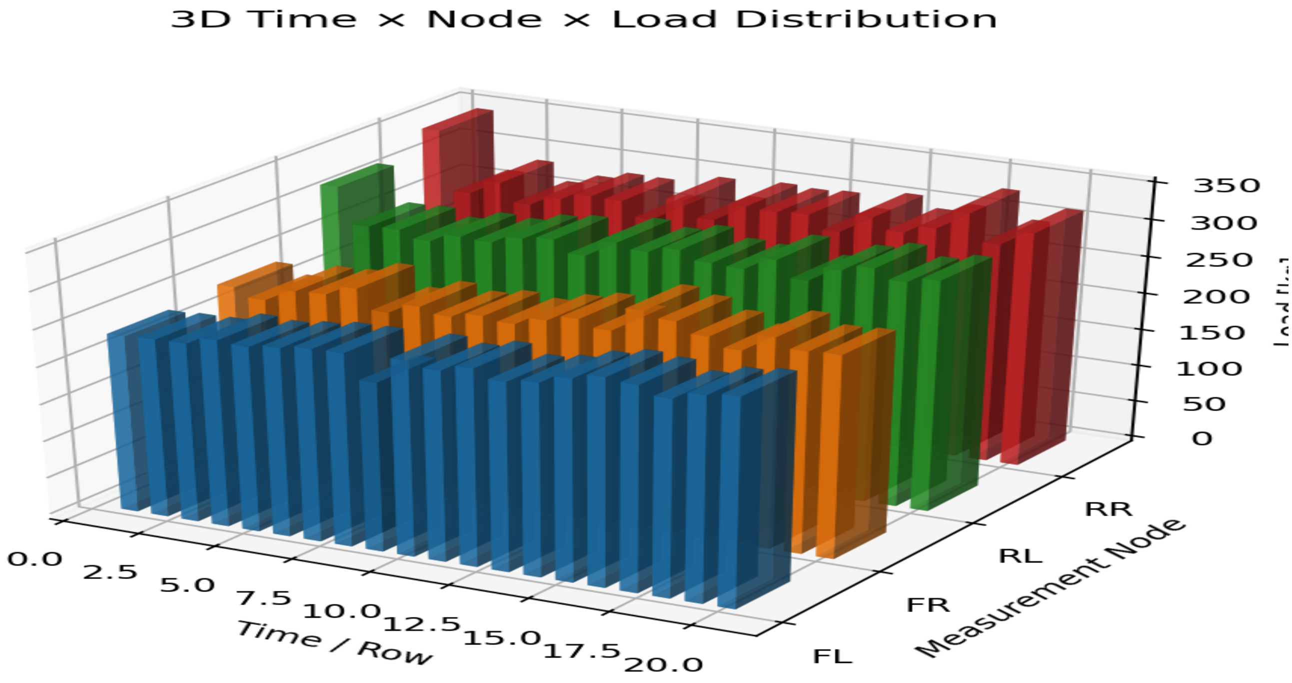
07 EVENT FREQUENCY



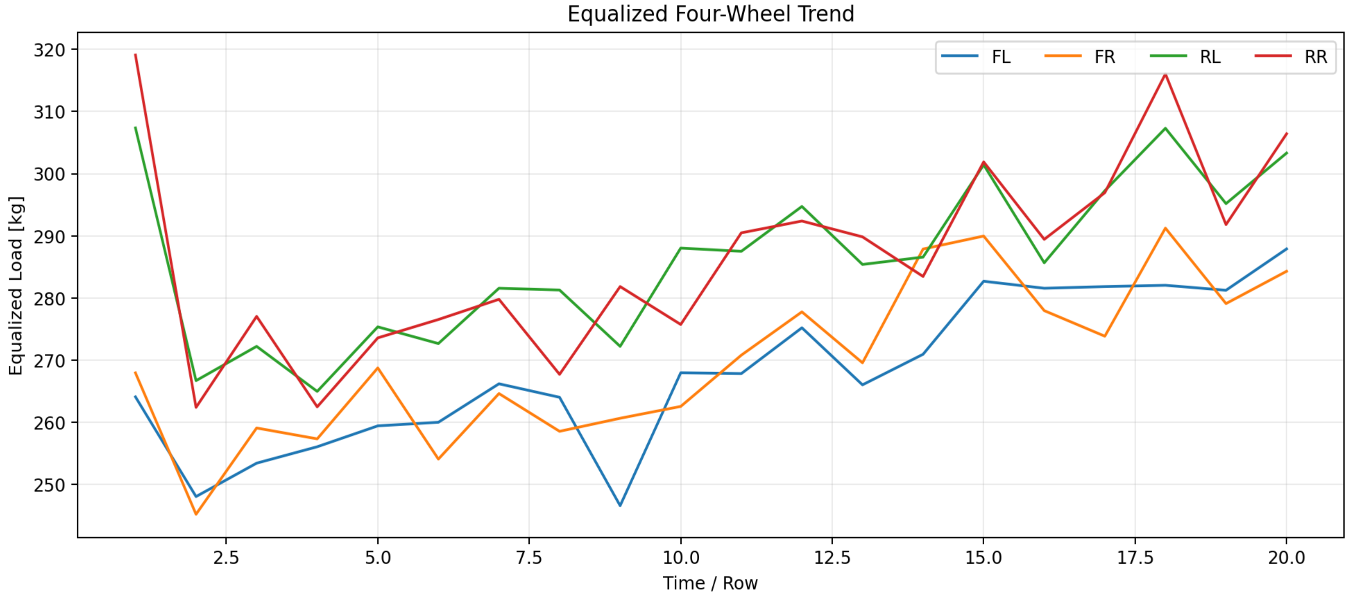
08 EQUALIZATION ADJUSTMENT



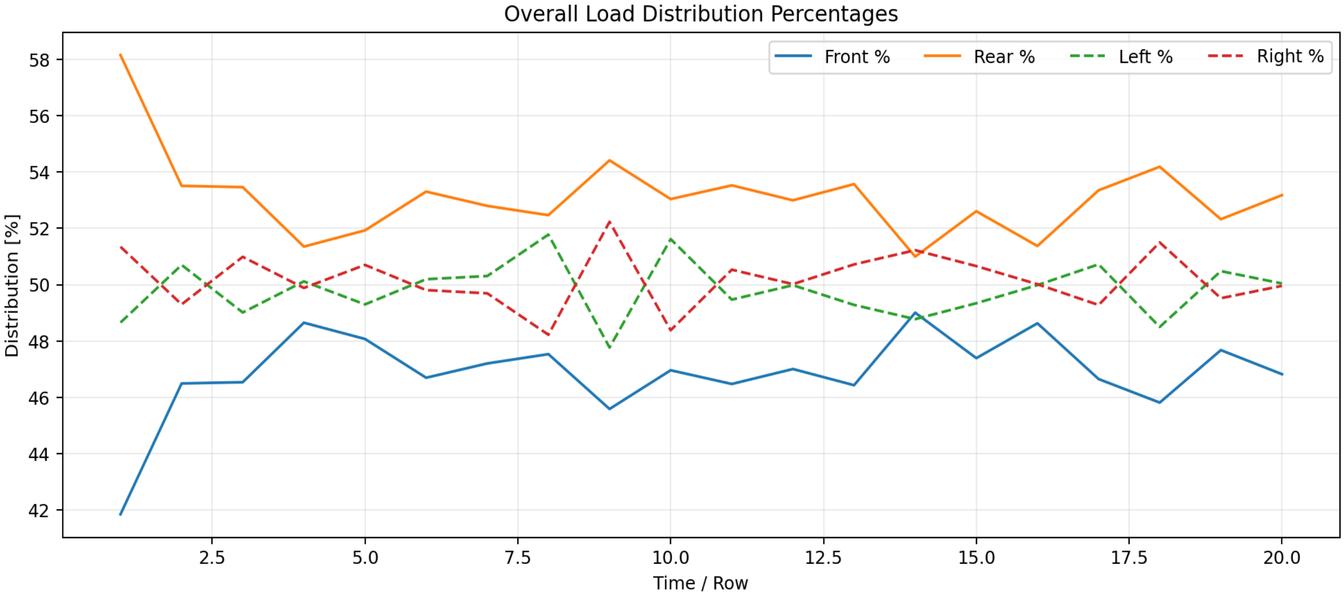
09 3D TIME NODE LOAD



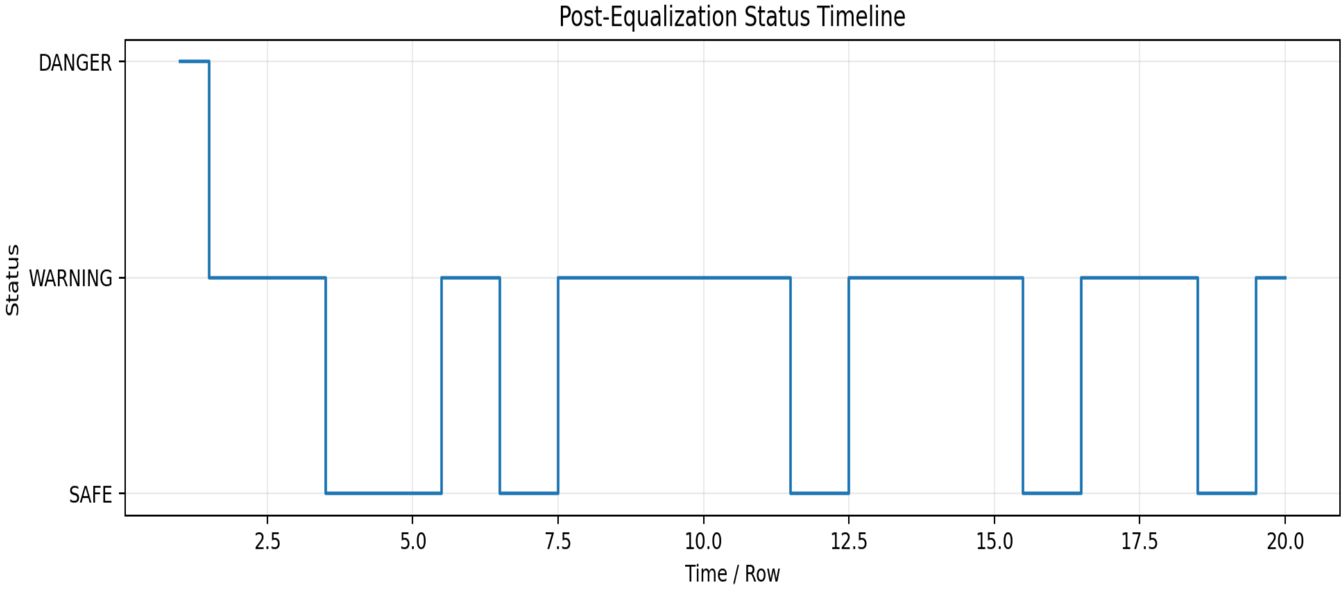
10 EQUALIZED FOUR WHEEL TREND



11 DISTRIBUTION PERCENTAGES



12 STATUS TIMELINE



HydroAI Engineering Insight

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Engineering note: the ±10 kg value is a project screening threshold. Manufacturer specifications, certified test procedures and applicable regulations remain authoritative.